Oscillation Control of a Double-Pendulum Overhead Crane based on Model-Matching Input Shaping Technique

Ho Duc Tho[1], Kazuhiko Terashima[2], and Takanori Miyoshi[1]

***Abstract*— In this paper, a model-matching input shaping controller will be designed for a double-pendulum overhead crane with time-varying cable length. It will be demonstrated that the propagation of vibration suppression allows us to eliminate oscillations for both pendulums, even though only one reference model is permitted. Comparative simulation results are presented to highlight the advantage of the proposed approach over the standard input shaping controller. An experimental study is also conducted on an actual crane.**

I. Introduction

Overhead cranes are primary machines for transporting heavy payloads, whose weights range from hundreds of kilograms to thousands of tons. The payload is suspended under a trolley by wire ropes like a pendulum, and the trolley moves horizontally along the rails to relocate the payload's position in the space. The rope length can be varied by a hoisting mechanism to adjust the payload's altitude to, for instance, avoid obstacles. During the trolley's maneuver, the payload exhibits a pendulum-like oscillation. Such a residual vibration adversely influences safety and productivity of the transportation. The circumstance even becomes much more complicated if the payload has a sizable dimension in length, because in this case, the double-pendulum effect must be taken into account. Due to its urgent needs, vibration suppression control for double-pendulum overhead cranes has extensively been investigated in recent years. These researches can be divided into two categories: feedback and feedforward controls. Feedback control techniques (e.g., energy and passivity-based control [1], [2], sliding mode control [3], [4], adaptive control [5], [6], model predictive control [7], nonlinear optimal control [8], [9]) are excellent in counteracting external disturbances. Nevertheless, they also require complex and costly sensor devices for measuring swing angles in real-time. On the other hand, feedforward controllers do not necessitate measurements of swing angles, i.e., the ability of eliminating the vibration is solely relied on the planned movement of the trolley. Therefore, in appropriate contexts where external disturbances are not harsh (i.e., in indoor environments), the feedforward control technique is preferred over the feedback one, as no additional sensor system is needed.

Input shaping control (see [10], [11]) remains to be an important technique in the feedforward control toolbox for dealing with vibratory systems due to its simplicity and effectiveness. The standard input shaping controllers perform exceptionally well for time-invariant cranes (i.e., when the rope length is constant) [12], [13] and even for multi-mode flexible systems like double-pendulum cranes [14], [15]. However, the standard input shaping controls applying on time-varying systems (e.g., cranes with varying cable length) often underperform because they are unable to render a complete vibration suppression. Until now, the common method to mitigate the adverse effect caused by time-varying dynamics is to use robust input shaper types such as zero derivatives reinforcement [10], extra insensitivity (EI) [11], [16], [17], and optimized specified insensitivity (SI) [18], [19]. A newer approach is to train a neural network to adjust magnitudes and time duration of the standard input shapers on the fly [20]. Nevertheless, these techniques are not capable of totally eliminating the vibration for time-varying crane systems, i.e., only a fraction of vibration can be reduced. Recently, a model reference input shaping controller has been proposed for a single-pendulum overhead crane with varying rope length that can perfectly suppress the residual vibration using a non-robust input shaper [21]. The basic idea of this technique is to match the actual time-varying crane with a (virtual) time-invariant model, on which the standard input shaping control can directly be applied. The (unshaped) reference command is then parameterized and solved to drive the trolley to the desired destination. We aim at expanding such an approach, which was designed for the single-pendulum crane, to the case of double-pendulum crane in this paper. It will later be shown that this extension is nontrivial, because the perfect model matching process cannot be performed for both swing angles. To resolve this problem, the propagation of vibration suppression principle will be proposed, namely that, if the swing angle of the second pendulum vanishes, then so it does for the first pendulum. Thus, it is only required to perform the perfect model matching on the second swing angle. To the best of our knowledge, there is not any existing input shaping-based controller that is able to completely suppress vibration for the double-pendulum crane with time-varying rope length.

II. Problem formulation

The mathematical model of a double-pendulum overhead crane with varying rope length is shown in Fig. 1. The trolley position, the first swing angle, the second swing angle, and the time-varying cable length are denoted as $x(t)$ (m), $\theta_1(t)$ (rad), $\theta_2(t)$ (rad), and $l_1(t)$ (m), respectively. The system parameters comprise m_1 (weight of the point-mass hook, in

[1]Ho Duc Tho and Takanori Miyoshi are with Department of System Safety, Nagaoka University of Technology, 1603-1 Kamitomiokamachi, Niigata, Japan. E-mails: hoductho@vos.nagaokaut.ac.jp, miyoshi@vos.nagaokaut.ac.jp

[2]Kazuhiko Terashima is with Toyohashi University of Technology, 1-1 Hibarigaoka, Aichi, Japan. E-mail: terasima@tut.ac.jp

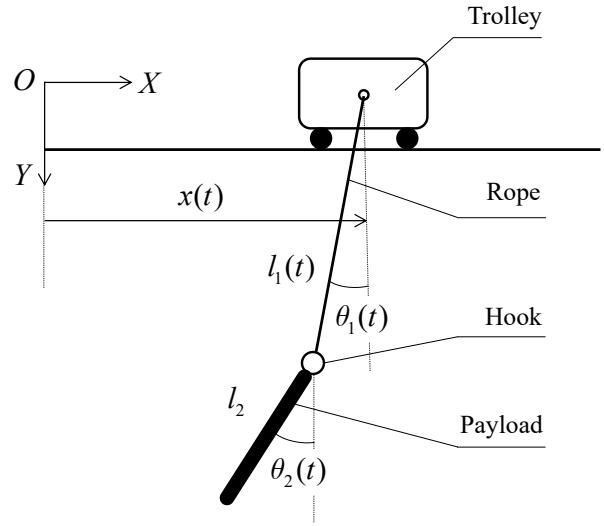

Fig. 1. A double-pendulum with time-varying cable length.

kg), m_2 (weight of the payload, in kg), J_2 (inertia of the payload, in kgm^2), l_2 (length of the payload, in m), and g (gravitational acceleration, $g = 9.81$ m/s^2). The Euler-Lagrange equation applying on the unactuated coordinates θ_1 and θ_2 results in the following nonlinear dynamics of the double-pendulum overhead crane [1]–[4]

$$\begin{aligned}&(m_1+m_2)\,l_1\ddot{\theta}_1+\frac{1}{2}m_2l_2\cos(\theta_1-\theta_2)\,\ddot{\theta}_2\\&\quad+2\,(m_1+m_2)\,\dot{l}_1\dot{\theta}_1+\frac{1}{2}m_2l_2\sin(\theta_1-\theta_2)\,\dot{\theta}_2^2\\&\qquad+(m_1+m_2)\,g\sin\theta_1=(m_1+m_2)\cos\theta_1\ddot{x}\end{aligned}\tag{1a}$$

$$\begin{aligned}&l_1\cos(\theta_1-\theta_2)\,\ddot{\theta}_1+\left(\frac{2J_2}{m_2l_2}+\frac{1}{2}l_2\right)\ddot{\theta}_2\\&\quad+2\cos(\theta_1-\theta_2)\,\dot{l}_1\dot{\theta}_1-l_1\sin(\theta_1-\theta_2)\,\dot{\theta}_1^2\\&\qquad+g\sin\theta_2=\cos\theta_2\ddot{x}-\sin(\theta_1-\theta_2)\,\ddot{l}_1.\end{aligned}\tag{1b}$$

Throughout the paper, it is supposed that the following assumptions hold

- The payload starts from rest [14], [15], [21], i.e.,
$$x(0)=\dot{x}(0)=\dot{l}_1(0)=\theta_i(0)=\dot{\theta}_i(0)=0,\ i=1,2.$$
- The swing angles θ_1 and θ_2 are sufficiently small so that the small-angle approximation is valid [13]–[15].
- The rope length function $l_1(t)$ is chosen such that $g-\ddot{l}_1(t)\neq 0\ \forall t\geq 0$.

The linearized model of the nonlinear crane (1a)–(1b) around its usual equilibrium point is

$$\begin{aligned}&l_1\ddot{\theta}_1+L_1\ddot{\theta}_2+2\dot{l}_1\dot{\theta}_1+g\theta_1=u\\&l_1\ddot{\theta}_1+L_2\ddot{\theta}_2+2\dot{l}_1\dot{\theta}_1+\ddot{l}_1\theta_1+(g-\ddot{l}_1)\theta_2=u,\end{aligned}\tag{2}$$

where $L_1 := 0.5m_2l_2/(m_1+m_2)$, $L_2 := 2J_2/(m_2l_2)+l_2/2$, and $u := \ddot{x}$ (the control input is the acceleration $\ddot{x}$ of the trolley). By using the linearized system (2), we wish to design an input shaping-based control law $u(t)$ so that at some finite *shaped* terminal time $\tau_f > 0$, the following terminal conditions should be realized

1) The trolley reaches its constant reference x_f at zero terminal speed, that is to say
$$x(\tau_f)=x_f,\ \dot{x}(\tau_f)=0.\tag{3}$$
2) To lift the payload, the rope is hoisted to a constant desired length $l_{1f}>0$ at zero terminal speed from an initial value $l_{10} := l_1(0) > 0$, namely that
$$l_1(\tau_f)=l_{1f},\ \dot{l}_1(\tau_f)=0.\tag{4}$$
3) Both swing angles vanish, i.e.,
$$\theta_1(\tau_f)=\dot{\theta}_1(\tau_f)=\theta_2(\tau_f)=\dot{\theta}_2(\tau_f)=0.\tag{5}$$

III. Model-matching input shaping control

We firstly formulate the propagation of vibration suppression principle in the following theorem.

Theorem 1. Consider the double-pendulum overhead crane system (2). Assume that there is $T>0$ so that $u(t)=0$ and $g-\ddot{l}_1(t)\neq 0\ \forall t\geq T$, then the following statements are true

1) If $\theta_1(t)=0\ \forall t\geq T$ then $\theta_2(t)=0\ \forall t\geq T$.
2) If $\theta_2(t)=0\ \forall t\geq T$ then $\theta_1(t)=0\ \forall t\geq T$.

Proof. We firstly prove 1). Since it is assumed that $\theta_1(t)=0\ \forall t\geq T$, we have $\dot{\theta}_1(t)=\ddot{\theta}_1(t)=0\ \forall t\geq T$. Substituting this result into the first equation of (2) gives $\ddot{\theta}_2(t)=0\ \forall t\geq T$. Thus, by the second equation of (2), it is true that $\theta_2(t)=0\ \forall t\geq T$ since it is supposed that $g-\ddot{l}_1(t)\neq 0\ \forall t\geq T$.

The statement 2) can also be proved in the same manner. Since $\theta_2(t)=0\ \forall t\geq T$, we have $\dot{\theta}_2(t)=\ddot{\theta}_2(t)=0\ \forall t\geq T$, for which (2) can be reduced to

$$\begin{aligned}&l_1\ddot{\theta}_1+2\dot{l}_1\dot{\theta}_1+g\theta_1=0\\&l_1\ddot{\theta}_1+2\dot{l}_1\dot{\theta}_1+\ddot{l}_1\theta_1=0.\end{aligned}$$

The above equations imply that $(\ddot{l}_1(t)-g)\theta_1(t)=0\ \forall t\geq T$, and hence, $\theta_1(t)=0\ \forall t\geq T$. This completes the proof. □

The model-matching input shaping controller will be designed on the basis of the propagation of vibration suppression principle. More specifically, the statement 2) of Theorem 1 will be employed, that is to say, the control input is devised to match the swing angle of the second pendulum with a reference oscillation. In order to do so, the model (2) needs to be decoupled to extract the relation between the second swing angle $\theta_2(t)$ and the control input $u(t)$ without being influenced by the first swing angle $\theta_1(t)$.

A. Decoupling dynamics

Denoting $M_1 := L_2 - L_1$ and $\alpha := g-\ddot{l}_1$. Note that $M_1>0$, and by assumption, $\alpha\neq 0$. Subtracting the first equation by the second one in (2) results in

$$\theta_1=\frac{M_1}{\alpha}\ddot{\theta}_2+\theta_2.\tag{6}$$

Differentiating (6) two times lead to the following results

$$\begin{aligned}\dot{\theta}_1&=\frac{M_1}{\alpha}\theta_2^{(3)}-\frac{M_1\dot{\alpha}}{\alpha^2}\ddot{\theta}_2+\dot{\theta}_2\\\ddot{\theta}_1&=\frac{M_1}{\alpha}\theta_2^{(4)}-\frac{2M_1\dot{\alpha}}{\alpha^2}\theta_2^{(3)}-\left[\frac{M_1\left(\alpha^2\ddot{\alpha}-2\alpha\dot{\alpha}^2\right)}{\alpha^4}-1\right]\ddot{\theta}_2,\end{aligned}\tag{7}$$

where $(.)^{(n)}$ denotes the n-th order derivative of the function $(.)$ with respect to time. Substituting (6)–(7) into the second equation of (2) yields

$$\theta_2^{(4)} + p_4(t)\theta_2^{(3)} + p_3(t)\ddot{\theta}_2 + p_2(t)\dot{\theta}_2 + p_1(t)\theta_2 = p_0(t)u, \tag{8}$$

in which the terms $p_i(t)$ $(i = 1, \ldots, 4)$ are given by

$$p_4(t) = -2\left(\frac{\dot{\alpha}}{\alpha} - \frac{\dot{l}_1}{l_1}\right)$$
$$p_3(t) = -\left[\frac{\ddot{\alpha}}{\alpha} + 2\frac{\dot{\alpha}}{\alpha}\left(\frac{\dot{l}_1}{l_1} - \frac{\dot{\alpha}}{\alpha}\right) - \frac{\alpha}{M_1}\left(1 + \frac{L_2}{l_1}\right) - \frac{\ddot{l}_1}{l_1}\right]$$
$$p_2(t) = \frac{2\alpha\dot{l}_1}{M_1 l_1},\ p_1(t) = \frac{g\alpha}{M_1 l_1},\ p_0(t) = \frac{\alpha}{M_1 l_1}.$$

The expression in (8) is the decoupling dynamics of the second swing angle $\theta_2(t)$ relative to the control input $u(t)$.

B. Control law design

By neglecting the time-varying functions related to the hoisting motion in (8), the following fourth-order reference model of the second swing angle is chosen

$$\xi_2^{(4)} + \lambda_2\ddot{\xi}_2 + \lambda_1\xi_2 = \lambda_0\bar{u}(t),\ \xi_2(0) = \ldots = \xi_2^{(3)}(0) = 0, \tag{9}$$

where $\xi_2(t)$ and $\bar{u}(t)$ denote the reference oscillation and the control input of the reference model, respectively. The terms λ_i $(i = 0, 1, 2)$ are given by

$$\lambda_2 = \frac{g}{M_1}\left(1 + \frac{L_2}{\bar{l}_1}\right),\ \lambda_1 = \frac{g^2}{M_1\bar{l}_1},\ \lambda_0 = \frac{g}{M_1\bar{l}_1},$$

in which $\bar{l}_1 > 0$ is a constant nominal value of the rope length, which can freely be chosen. It is straightforward to prove that the reference system (9) always has two distinct vibration modes at the natural frequencies $\omega_1, \omega_2 > 0$ $(\omega_1 \neq \omega_2)$, which are determined by taking the square root of the absolute value of the two different roots associating with the characteristic equation $s^2 + \lambda_2 s + \lambda_1 = 0$ (s is the complex variable). Let $T_1 := 2\pi/\omega_1$ and $T_2 := 2\pi/\omega_2$ be natural periods corresponding to the natural frequencies ω_1 and ω_2, respectively. Denoting $\star$ as the convolution operator and $\delta(.)$ as the Direct delta distribution, the reference control input $\bar{u}(t)$ of the multi-mode system (9) is a standard input shaping controller, as described as follows [14], [22]

$$\begin{aligned}
\bar{u}(t) &= u_r(t) \star I_{\omega 1}(t) \star I_{\omega 2}(t) \\
I_{\omega 1}(t) &= 0.5\left[\delta(t) + \delta\left(t - \frac{T_1}{2}\right)\right] \\
I_{\omega 2}(t) &= 0.5\left[\delta(t) + \delta\left(t - \frac{T_2}{2}\right)\right].
\end{aligned} \tag{10}$$

The *unshaped* reference command $u_r(t)$ is parameterized by

$$u_r(t) = \begin{cases} \beta_1 t + \beta_0, & t \in [0,\ t_f] \\ 0, & t > t_f, \end{cases} \tag{11}$$

where t_f—the *unshaped* terminal time—is a positive real number chosen by the designer, and β_i $(i = 0, 1)$ are to-be-determined variables to realize the control objective (3). When the control input (10) is applied on the reference model (9), the reference oscillation $\xi_2(t)$ is suppressed at the (shaped) terminal time $\tau_f = t_f + T_1/2 + T_2/2$, namely that, $\xi_2(t) = \ldots = \xi_2^{(4)}(t) = 0\ \forall t \geq \tau_f$. We then proceed to perform the perfect model matching to match the second swing angle $\theta_2(t)$ with its reference $\xi_2(t)$. From (8) and (9), it can be obtained that

$$\theta_2^{(4)} = -\left[p_4(t)\theta_2^{(3)} + p_3(t)\ddot{\theta}_2 + p_2(t)\dot{\theta}_2 + p_1(t)\theta_2 - p_0(t)u\right]$$
$$\xi_2^{(4)} = -\left[\lambda_2\ddot{\xi}_2 + \lambda_1\xi_2 - \lambda_0\bar{u}\right].$$

By letting $\theta_2^{(i)}(t) \equiv \xi_2^{(i)}(t)$ $(i = 0, \ldots, 4)$, the following constraint needs to be kept

$$\begin{aligned}
&p_4(t)\xi_2^{(3)} + p_3(t)\ddot{\xi}_2 + p_2(t)\dot{\xi}_2 + p_1(t)\xi_2 - p_0(t)u \\
&\quad = \lambda_2\ddot{\xi}_2 + \lambda_1\xi_2 - \lambda_0\bar{u}.
\end{aligned}$$

From the above equation, the model-matching input shaping control law can be determined by

$$u(t) = \frac{\sum_{i=0}^{3} p_{i+1}(t)\xi_2^{(i)}(t) - \sum_{i=0}^{1}\lambda_{2-i}\xi_2^{(-2i+2)}(t) + \lambda_0\bar{u}(t)}{p_0(t)}. \tag{12}$$

By employing the control law (12), the real oscillation is completely suppressed after the terminal time τ_f. For this reason, the control objective (5) is realized.

Next, to fulfill the control objective (4), the cable length function $l_1(t)$ is chosen to be a third-order time-dependent polynomial as follows

$$l_1(t) = \sum_{i=0}^{3}\alpha_i t^i = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3, \tag{13}$$

where $\alpha_0 = l_{10}$, $\alpha_1 = 0$, $\alpha_2 = 3\left(l_{1f} - l_{10}\right)/\tau_f^2$, and $\alpha_3 = -2\left(l_{1f} - l_{10}\right)/\tau_f^3$. As of now, both trolley's control input $u(t)$ and the hoisting motion $l_1(t)$ are designed to respond to the control objectives (4)–(5). In the subsequent computations, the unknown variables β_0 and β_1 of the unshaped reference command $u_r(t)$ (see (11)) will be analytically determined to realize the remaining control objective (3) with regard to the trolley's terminal destination and speed. The first step to achieve this requirement is to compute the closed-form solution of the second swing angle $\theta_2(t)$, or equivalently, the reference oscillation $\xi_2(t)$. The Laplace transformation technique will be employed. Denote $\mathcal{L}$ as the Laplace transform operator and recall that s is the complex variable. From (11), the Laplace transform $\mathcal{U}_r(s)$ of the unshaped reference command $u_r(t)$ is

$$\begin{aligned}
&\mathcal{U}_r(s) = \mathcal{L}\left\{u_r(t)\right\} = \int_0^{t_f}\left(\beta_0 + \beta_1 t\right)e^{-st}dt \\
&\Longrightarrow \mathcal{U}_r(s) = \frac{\beta_0}{s} + \frac{\beta_1}{s^2} - \left(\frac{\beta_0 + \beta_1 t_f}{s} + \frac{\beta_1}{s^2}\right)e^{-st_f}.
\end{aligned} \tag{14}$$

Moreover, the Laplace transforms of the input shapers $I_{\omega 1}(t)$ and $I_{\omega 2}(t)$ are $\mathcal{I}_{\omega 1}(s) = 0.5\left(1+e^{-sT_1/2}\right)$ and $\mathcal{I}_{\omega 2}(s) = 0.5\left(1+e^{-sT_2/2}\right)$, respectively. Applying the Laplace operator on the reference model (9) yields

$$\Xi_2(s) = \frac{\lambda_0 \mathcal{U}_r(s)\mathcal{I}_{\omega 1}(s)\mathcal{I}_{\omega 2}(s)}{s^4+\lambda_2 s^2+\lambda_1}, \tag{15}$$

where $\Xi_2(s)$ is the Laplace transform of $\xi_2(t)$ and the result $\mathcal{L}\left\{u_r(t)\star I_{\omega 1}(t)\star I_{\omega 2}(t)\right\} = \mathcal{U}_r(s)\mathcal{I}_{\omega 1}(s)\mathcal{I}_{\omega 2}(s)$ has been used. By substituting (14) into (15) and rearranging the result, it can be obtained that

$$\begin{aligned}\Xi_2(s) =& \frac{\lambda_0}{4}\left[\mathcal{F}_1(s)-\mathcal{F}_2(s)e^{-st_f}\right]\times \\ & \left[1+e^{-sT_1/2}+e^{-sT_2/2}+e^{-s(T_1+T_2)/2}\right],\end{aligned} \tag{16}$$

in which $\mathcal{F}_i(s)$ and their inverse Laplace transforms $f_i(t)$ $(i=1,2)$ are given as follows

$$\begin{aligned}\mathcal{F}_1(s) &= \frac{\beta_0 s^{-1}+\beta_1 s^{-2}}{\left(s^2+\omega_1^2\right)\left(s^2+\omega_2^2\right)}, \\ \mathcal{F}_2(s) &= \frac{\left(\beta_0+\beta_1 t_f\right)s^{-1}+\beta_1 s^{-2}}{\left(s^2+\omega_1^2\right)\left(s^2+\omega_2^2\right)}, \\ f_1(t) &= \beta_0\left(\frac{1}{\Omega_\times^2}+\frac{\cos\omega_1 t}{\omega_1^2\Omega_{2-}}+\frac{\cos\omega_2 t}{\omega_2^2\Omega_{2-}}\right) \\ &\quad +\beta_1\left(\frac{t}{\Omega_\times^2}+\frac{\sin\omega_1 t}{\omega_1^3\Omega_{2-}}+\frac{\sin\omega_2 t}{\omega_2^3\Omega_{2-}}\right), \\ f_2(t) &= \beta_0\left(\frac{1}{\Omega_\times^2}+\frac{\cos\omega_1 t}{\omega_1^2\Omega_{2-}}-\frac{\cos\omega_2 t}{\omega_2^2\Omega_{2-}}\right) \\ &\quad +\beta_1\left(\frac{t+t_f}{\Omega_\times^2}+\frac{\sin\omega_1 t+\omega_1 t_f\cos\omega_1 t}{\omega_1^3\Omega_{2-}}\right. \\ &\qquad \left.-\frac{\sin\omega_2 t+\omega_2 t_f\cos\omega_2 t}{\omega_2^3\Omega_{2-}}\right),\end{aligned} \tag{17}$$

where it is defined that $\Omega_\times := \omega_1\omega_2$, and $\Omega_{2-} := \omega_1^2-\omega_2^2$. Taking the inverse Laplace transform of (16), we have

$$\begin{aligned}\xi_2(t) = \frac{\lambda_0}{4}\Bigg[& f_1(t)-\sum_{\sigma\in\{t,\tau\}} f_2\left(t-\sigma_f\right)H\left(t-\sigma_f\right) \\ & +\sum_{i=1}^{2} f_1\left(t-\frac{T_i}{2}\right)H\left(t-\frac{T_i}{2}\right) \\ & -\sum_{i=1}^{2} f_2\left(t-t_f-\frac{T_i}{2}\right)H\left(t-t_f-\frac{T_i}{2}\right) \\ & +f_1\left(t-\frac{T_1}{2}-\frac{T_2}{2}\right)H\left(t-\frac{T_1}{2}-\frac{T_2}{2}\right)\Bigg],\end{aligned} \tag{18}$$

where $H(.)$ is the Heaviside function. The analytical solution of the reference oscillation can now be obtained in (18). We then proceed to integrate the system dynamics (8) to determine the position $x(t)$ and the speed $\dot{x}(t)$ of the trolley. Denote $h_4(t) := 1/p_0(t)$, $h_3(t) := p_4(t)/p_0(t)$, $h_2(t) := p_3(t)/p_0(t)$, $h_1(t) := p_2(t)/p_0(t)$, and $h_0(t) := 1/p_0(t)$, the decoupled dynamics (8) can be rewritten as

$$h_4(t)\xi_2^{(4)}+h_3(t)\xi_2^{(3)}+h_2(t)\ddot{\xi}_2+h_1(t)\dot{\xi}_2+h_0(t)\xi_2 = \ddot{x}. \tag{19}$$

The result of the integration process performed on (19) is summarized as follows

$$\begin{aligned}\dot{x}(t) &= h_4(t)\xi_2^{(3)}(t)+\dot{h}_4(t)\ddot{\xi}_2(t)+\left[L_2+l_1(t)\right]\dot{\xi}_2(t) \\ &\quad +\dot{l}_1(t)\xi_2(t)-(2\alpha_2-g)\int_0^t \xi_2 d\tau-6\alpha_3\int_0^t \tau\xi_2 d\tau \\ x(t) &= h_4(t)\ddot{\xi}_2(t)+\left[L_2+l_1(t)\right]\xi_2(t)+6\alpha_3\int_0^t \tau^2\xi_2 d\tau \\ &\quad -(2\alpha_2-g)\,t\int_0^t \xi_2 d\tau+(2\alpha_2-6\alpha_3 t-g)\int_0^t \tau\xi_2 d\tau.\end{aligned} \tag{20}$$

By applying the control objective (3) on (20) and noting that $\xi_2(\tau_f) = \ldots = \xi_2^{(3)}(\tau_f) = 0$, the following result holds

$$\begin{aligned}(2\alpha_2-g)\int_0^{\tau_f}\xi_2(t)dt+6\alpha_3\int_0^{\tau_f}t\xi_2(t)dt &= 0 \\ (2\alpha_2-g)\int_0^{\tau_f}t\xi_2(t)dt+6\alpha_3\int_0^{\tau_f}t^2\xi_2(t)dt &= x_f.\end{aligned} \tag{21}$$

The integral terms in (21) will be evaluated next. By using the expression of $\xi_2(t)$ in (18), it is true that

$$\int_0^{\tau_f} t^i\xi_2(t)dt = \Phi_i\beta_0+\Gamma_i\beta_1,\ i=0,1,2, \tag{22}$$

where, by denoting $\Omega_+ := \omega_1+\omega_2$, $\Omega_{2+} := \omega_1^2+\omega_2^2$, $\varphi := \lambda_0 t_f/6$, and $\gamma := \lambda_0 t_f^2/12$, the terms Φ_i and Γ_i $(i=0,1,2)$ are given by

$$\begin{aligned}\Phi_0 &= \frac{\lambda_0 t_f}{\Omega_\times^2},\ \Gamma_0 = \frac{\lambda_0 t_f^2}{2\Omega_\times^2} \\ \Phi_1 &= \frac{\lambda_0 t_f\left(\pi\Omega_+ + t_f\Omega_\times\right)}{2\Omega_\times^3},\ \Gamma_1 = \frac{\lambda_0 t_f^2\left(3\pi\Omega_+ + 4t_f\Omega_\times\right)}{12\Omega_\times^3} \\ \Phi_2 &= \frac{\varphi\left[3\left(\pi^2-4\right)\Omega_{2+}+\Omega_\times(3\pi t_f\Omega_+ + 2t_f^2\Omega_\times + 3\pi^2)\right]}{\Omega_\times^4} \\ \Gamma_2 &= \frac{\gamma\left[3\left(\pi^2-4\right)\Omega_{2+}+\Omega_\times(4\pi t_f\Omega_+ + 3t_f^2\Omega_\times + 3\pi^2)\right]}{\Omega_\times^4}.\end{aligned} \tag{23}$$

Substituting the result in (22) into (21) yields

$$\begin{aligned}&\boldsymbol{M}\begin{bmatrix}\beta_0\\ \beta_1\end{bmatrix} = \begin{bmatrix}0\\ x_f\end{bmatrix},\ \text{where} \\ &\boldsymbol{M} := \begin{bmatrix}(2\alpha_2-g)\Phi_0+6\alpha_3\Phi_1 & (2\alpha_2-g)\Gamma_0+6\alpha_3\Gamma_1 \\ (2\alpha_2-g)\Phi_1+6\alpha_3\Phi_2 & (2\alpha_2-g)\Gamma_1+6\alpha_3\Gamma_2\end{bmatrix}.\end{aligned} \tag{24}$$

From (24), the unknown variables β_0 and β_1 of the unshaped reference command $u_r(t)$ can be determined by

$$\begin{bmatrix}\beta_0\\ \beta_1\end{bmatrix} = \boldsymbol{M}^{-1}\begin{bmatrix}0\\ x_f\end{bmatrix}. \tag{25}$$

It can be seen that the solvability of the unknowns β_0 and β_1 depends on the invertibility of the matrix $\boldsymbol{M}$. It will be shown that, in some cases, the matrix $\boldsymbol{M}$ is singular and the

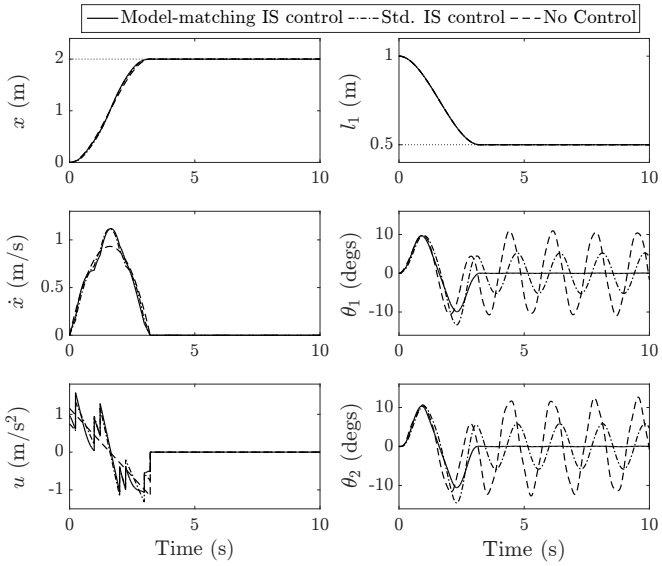

Fig. 2. Comparative simulation result (versus the standard input shaping control and no-control cases).

solution of β_0 and β_1 cannot be (uniquely or consistently) defined. However, we can avoid these singularities by simply changing the value of the unshaped terminal time t_f.

Theorem 2. There is a finite number of unshaped terminal times t_f (which can freely be chosen by the designer) that makes the matrix $\boldsymbol{M}$ defined in (24) singular.

Proof. From (23), the terms Φ_i and Γ_i $(i = 0, 1, 2)$ are polynomials with respect to the unshaped terminal time t_f. Moreover, the terms α_2 and α_3 are also polynomial fractions with respect to t_f (see (13)). Therefore, the determinant of the matrix $\boldsymbol{M}$, denoted as $\det(\boldsymbol{M})$, is also a polynomial fraction of t_f. For this reason, the equation $\det(\boldsymbol{M}) = 0$ has a finite number of isolated roots in terms of t_f, as a result of the fundamental theorem of algebra [23, Thereom D.4]. By noting that the matrix $\boldsymbol{M}$ is singular if and only if $\det(\boldsymbol{M}) = 0$, the proof can now be completed. □

An immediate consequence of Theorem 2 is that the solvability of the unknowns β_0 and β_1 can always be ensured since the unshaped terminal time t_f can arbitrarily be chosen by the designer. By this, the design process of the model-reference input shaping controller is therefore accomplished.

IV. Simulation result

In this section, the efficacy of the proposed model-matching input shaping controller will be verified. Moreover, a comparative simulation study will be conducted to compare the proposed approach with a no-control case and the standard input shaping controller. The rope is varied from an initial length of $l_{10} = 1$ m to a terminal length of $l_{1f} = 0.5$ m. The constant reference value for the trolley position is $x_f = 2$ m. The nominal rope length of the reference model (9) is chosen as $\bar{l}_1 = 0.5\,(l_{10} + l_{1f})$. The unshaped terminal time t_f is fixed at $t_f = 2$ s. The length, the mass, and the inertia of the payload are $l_2 = 0.4$ m, $m_2 = 4.99$ kg, and $J_2 = 0.065$ kgm^2, respectively. Based on (13) and (25), essential parameters for the rope length function $l_1(t)$ and the unshaped reference command $u_r(t)$ can be determined as $\alpha_0 = 1$, $\alpha_1 = 0$, $\alpha_2 = -0.1447$, $\alpha_3 = 0.0299$, and $\beta_0 = 2.9821$, $\beta_1 = -3.004$, respectively. The computational time is 0.0073 s on a personal computer with 2.7-GHz Intel Core i5-4310M using MATLAB, indicating that a real-time implementation is possible.

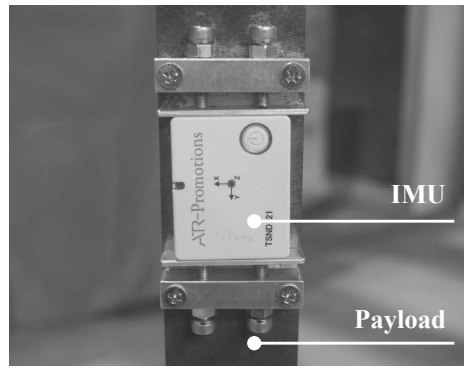

Fig. 3. The IMU is attached on the payload in order to measure the second swing angle.

In the no-control context, the unshaped reference command $u_r(t)$ profile shown in (11) is employed to drive the trolley to its reference x_f without considering the vibration suppression. Moreover, in the case of standard input shaping control scheme, the reference control input $\bar{u}(t)$ in (10) is directly applied for the trolley motion, i.e., $u(t) = \bar{u}(t)$ is utilized. An identical rope length function $l_1(t)$ (as previously computed) is used for both no-control and standard input shaping control cases.

It is worth emphasizing that the transfer times of the three methods (i.e., the proposed model-matching input shaping control law (12), the no-control case, and the standard input shaping scheme) are forced to be the same in order for the comparison to be fair. As a result, performance of each approach may entirely be assessed by the amount of residual vibration at the completion of transportation.

The comparative simulation result is depicted in Fig. 2. It is remarked that the full time-varying nonlinear system (1a)–(1b) is employed in all simulations. It can be observed that the proposed controller is able to completely eliminate the residual vibration at the end of the maneuver. In contrast, a substantial vibration amount persists in the case of standard input shaping control and no-control scenarios.

V. Experimental result

The proposed model-matching input shaping controller is now validated on a small-scale overhead crane. The system hardware and components are detailed in [24]. In addition, to measure the second swing angle, an inertial measurement unit (IMU) TSND121 of ATR Promotions is attached on the payload (see Fig. 3). Due to dimensional limitations, the reference value of the trolley position is fixed at $x_f = 1$ m. Moreover, the rope is hoisted from an initial length of

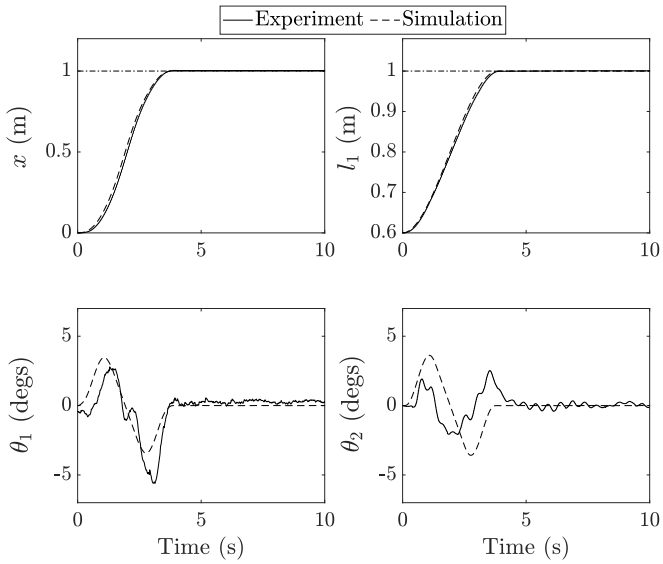

Fig. 4. Experimental result of the model-matching input shaping controller.

$l_{10} = 0.6$ m to a terminal length of $l_{1f} = 1$ m. The shaped terminal time is chosen as $\tau_f = 3.85$ s. The remaining system parameters are similar to those described in section IV. Based on (13) and (25), essential parameters for the rope length function $l_1(t)$ and the unshaped reference command $u_r(t)$ can be determined as $\alpha_0 = 0.6$, $\alpha_1 = 0$, $\alpha_2 = 0.0812$, $\alpha_3 = -0.0141$ and $\beta_0 = 1.038$, $\beta_1 = -0.859$, respectively.

The experimental result is shown in Fig. 4. It can be seen that the trolley precisely arrives at the destination while both swing angles are almost completely suppressed. Moreover, the experimental result resembles the simulation result well. This therefore confirms the effectiveness of the proposed model-matching input shaping controller.

VI. Conclusions and Future Work

The model-matching input shaping control technique has been extended to the case of double-pendulum overhead crane system with time-varying rope length. The propagation of vibration suppression principle was introduced to enable complete vibration elimination for both swing angles, even though the exact model matching can only be performed on the second swing angle. Through simulation results, the proposed approach demonstrated its superiority over the standard input shaping controller. The experimental result confirmed the effectiveness of the proposed method. In our future work, we aim at applying the established model-matching input shaping controller on a real-size crane to verify its practicability.

References

[1] N. Sun, T. Yang, Y. Fang, Y. Wu, and H. Chen, "Transportation control of double-pendulum cranes with a nonlinear quasi-pid scheme: Design and experiments," *IEEE Trans. on Systems, Man, and Cybernetics: Systems*, vol. 49, no. 7, pp. 1408–1418, 2019.

[2] D. Qian, *Anti-Sway Control for Cranes: Design and Implementation Using MATLAB*, 1st ed. Walter de Gruyter GmbH, 2018.

[3] L. A. Tuan and S.-G. Lee, "Sliding mode controls of double-pendulum crane systems," *J. of Mechanical Science and Technology*, vol. 27, no. 6, pp. 1863–1873, Jun 2013.

[4] H. Ouyang, J. Hu, G. Zhang, L. Mei, and X. Deng, "Sliding-mode-based trajectory tracking and load sway suppression control for double-pendulum overhead cranes," *IEEE Access*, vol. 7, pp. 4371–4379, 2019.

[5] M. Zhang, X. Ma, X. Rong, X. Tian, and Y. Li, "Adaptive tracking control for double-pendulum overhead cranes subject to tracking error limitation, parametric uncertainties and external disturbances," *Mechanical Systems and Signal Processing*, vol. 76-77, pp. 15–32, 2016.

[6] B. Lu, Y. Fang, and N. Sun, "Enhanced-coupling adaptive control for double-pendulum overhead cranes with payload hoisting and lowering," *Automatica*, vol. 101, pp. 241–251, 2019.

[7] M. Thomas and O. Sawodny, "Flatness-based feedforward and modal model-predictive state-feedback control of a double pendulum bridge crane," *IFAC-PapersOnLine*, vol. 55, no. 27, pp. 19–24, 2022, iFAC Symp. on Mechatronic Systems.

[8] G. Rigatos, "Nonlinear optimal control for the underactuated double-pendulum overhead crane," *J. of Vibration Engineering & Technologies*, Mar 2023.

[9] M. Li, H. Chen, and Z. Li, "Input-limited optimal control for overhead cranes with payload hoisting/lowering and double pendulum effects," *Nonlinear Dynamics*, vol. 111, no. 12, pp. 11 135–11 151, Jun 2023.

[10] N. C. Singer and W. P. Seering, "Preshaping command inputs to reduce system vibration," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 112, no. 1, pp. 76–82, Mar 1990.

[11] W. Singhose, "Command shaping for flexible systems: A review of the first 50 years," *Int. J. of Precision Engineering and Manufacturing*, vol. 10, no. 4, pp. 153–168, Oct 2009.

[12] M. F. Daqaq and Z. N. Masoud, "Nonlinear input-shaping controller for quay-side container cranes," *Nonlinear Dynamics*, vol. 45, no. 1, pp. 149–170, Jul 2006.

[13] A. Wahrburg, J. Jurvanen, M. Niemelä, and M. Holmberg, "Input shaping for non-zero initial conditions and arbitrary input signals with an application to overhead crane control," in *IEEE Int. Conf. on Advanced Motion Control*, 2022, pp. 36–41.

[14] W. Singhose, D. Kim, and M. Kenison, "Input shaping control of double-pendulum bridge crane oscillations," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 130, no. 3, p. 034504, 05 2008.

[15] Z. N. Masoud and K. A. Alhazza, "Frequency-modulation input shaping control of double-pendulum overhead cranes," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 136, no. 2, p. 021005, 12 2013.

[16] J. Vaughan, A. Yano, and W. Singhose, "Robust negative input shapers for vibration suppression," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 131, no. 3, Mar 2009.

[17] K. H. Rew, C. W. Ha, and K. S. Kim, "An impulse-time perturbation approach for enhancing the robustness of extra-insensitive input shapers," *Automatica*, vol. 49, no. 11, pp. 3425–3431, 2013.

[18] N. Singer and W. Seering, "An extension of command shaping methods for controlling residual vibration using frequency sampling," in *Proc. of IEEE Int. Conf. on Robotics and Automation*, vol. 1, 1992, pp. 800–805.

[19] L. Y. Pao and M. A. Lau, "Robust input shaper control design for parameter variations in flexible structures," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 122, no. 1, pp. 63–70, Oct 1999.

[20] L. Ramli, Z. Mohamed, and H. I. Jaafar, "A neural network-based input shaping for swing suppression of an overhead crane under payload hoisting and mass variations," *Mechanical Systems and Signal Processing*, vol. 107, pp. 484 – 501, 2018.

[21] H. D. Tho, K. Terashima, and T. Miyoshi, "Vibration control of an overhead crane with hoisting motion using input shaping technique," in *American Control Conf.*, 2022, pp. 1910–1914.

[22] J. Shan, H.-T. Liu, and D. Sun, "Modified input shaping for a rotating single-link flexible manipulator," *Journal of Sound and Vibration*, vol. 285, no. 1, pp. 187–207, 2005.

[23] D. Poole, *Linear Algebra: A Modern Introduction*, 4th ed. Cengage Learning, 2015.

[24] H. D. Tho, N. Uchiyama, and K. Terashima, "Resonance-based tossing control for bulk materials transportation of an overhead crane," *IEEE Trans. on Industrial Electronics*, vol. 68, no. 1, pp. 609–621, 2021.